

0. Critique of the brief

Author's critique: The brief correctly rejects a binary review, but its title and requested policy inference are stronger than the corpus, and it omits several distinctions needed to make the answer auditable.

Primary critique

- The title's "pros and cons" language invites the binary organization that the brief correctly rules out; the review should treat the title as a policy question and organize the evidence around the institutional sequence through which an earmark could affect revenue and expenditure.
- "A solution to domestic revenue mobilization" is stronger than the evidence permits; the corpus contains evidence on tax passage, stated support, gross receipts, statutory dedication, sectoral expenditure, and service use, but almost no identified evidence on the net effect of earmarking on aggregate domestic revenue in SSA.
- The outcome must be decomposed into tax enactment, gross collection, net general-government revenue, target-sector additionality, expenditure composition, expenditure quality, and service outcomes; treating any one of these as "DRM" would obscure several possible offsetting effects.

Supporting critique

- The merged file combines faithful paper summaries with project-layer interpretations; the review uses the faithful summaries as evidence and reserves cross-paper interpretation for Section 11.
- Several entries concern earmarked grants, budget support, general tax morale, spending shocks, or optimal taxation rather than earmarked taxes; they are identified as adjacent evidence rather than silently transferred to the core object.
- The direct SSA evidence is concentrated in Ghana, African road funds, selected health-financing cases, and a small number of descriptive institutional observations; an SSA-wide prior can be motivated but not established from this corpus.
- Page references marked "~" reproduce approximate anchors in the summary bank and require source verification before final publication.
- Akitoby et al.'s general DRM material and Chelsky's general budget-support discussion belong in the introduction and are excluded here; Chelsky is retained only where tax-policy reversal is directly relevant.

0.1 What the candidate structures miss

The candidate spines omit legal status, marginality, expenditure quality, durability, and budget-wide control, so neither can carry the full earmarking-to-DRM argument.

Primary omissions

- "Enactment → compliance → execution → fiscal cost" is too linear because execution contains at least three distinct stages: revenue transfer, incremental sectoral expenditure, and conversion of expenditure into usable outputs.
- "Enactment → compliance → execution → fiscal cost" treats budget fragmentation, cash management, loss of comprehensiveness, and contingent liabilities as terminal costs rather than system conditions that affect every earlier stage.
- The five-channel structure—passage, compliance, commitment, trust, and benefit link—mixes causal mechanisms with outcomes and allows the same paper to appear repeatedly without clarifying which fiscal stage it actually informs.

Supporting omissions

- Neither proposed structure begins with the legal object being coded, although the distinction between a statutory rule, an implementing regulation, an appropriation, a transfer, and an executed expenditure is central to the proposed empirical contribution.
- Neither proposed structure gives expenditure quality and fund governance equal standing with revenue attribution.
- Neither proposed structure contains an explicit durability stage or a comparison between repeal, suspension, diversion, and formal amendment.
- Neither proposed structure accommodates resource-revenue funds, social contributions, earmarked grants, and user charges as boundary cases rather than interchangeable forms of earmarking.

1. Proposed spine and treatment of the corpus

Author's structural finding: A transmission chain is the only proposed spine that preserves the relevant legal, political, revenue, budget, expenditure, and system-wide distinctions without turning pro/con or theory/empirics into competing literatures.

1.1 Proposed spine

The chosen spine treats earmarking as a sequence of institutional gates from legal authorization through budget-wide fiscal control.

Primary structure

- Section 2 defines the legal and fiscal object, establishes the boundary of the sample, and defends de jure coding as the first—not final—object of measurement.
- Section 3 covers political authorization: passage, coalition formation, consent, salience, and discriminatory political support.
- Section 4 covers revenue realization: compliance, gross yield, tax-base effects, aggregate revenue substitution, and policy durability.
- Section 5 covers budget translation: transfer, release, execution, marginality, additionality, fungibility, and expenditure composition.
- Section 6 covers expenditure quality and fund governance: procurement, absorptive capacity, managerial incentives, audit, accountability, and service delivery.

Supporting structure

- Section 7 treats earmarking as a budget-comprehensiveness and extrabudgetary-funds problem.
- Section 8 records the institutional conditions and SSA evidence that shift the prior before any causal estimation.
- Section 9 states what the corpus does not establish.
- Section 10 supplies the proposed evidence table.
- Section 11 derives the channels and conditions as the author's synthesis and bridges to the conceptual framework.

1.2 Rejected alternatives

The rejected alternatives collapse mechanisms, outcomes, sectors, or legal attributes that the corpus treats separately.

Primary rationale

- A pro/con spine is rejected because Buchanan, Teja, Lee and Wagner, McCleary, Cashin et al., and Bird and Jun all make the sign contingent on different assumptions, institutional arrangements, or counterfactuals.
- A theory-versus-empirics spine is rejected because the relevant gap is usually between a theoretical mechanism and evidence at a particular fiscal stage, not between two self-contained literatures.
- A sectoral spine—health, roads, education, environment—is rejected because it would hide common mechanisms such as political packaging, fungibility, fund governance, and off-budget fragmentation.

Supporting rationale

- A hard-versus-soft spine is rejected because the corpus does not establish a general performance ranking and because "hardness" can refer separately to enactment law, transfer rules, appropriation discretion, amendment authority, or actual execution.

1.3 Corpus treatment

The corpus must be stratified by direct relevance, instrument, setting, and evidentiary function to prevent adjacent findings from being read as direct evidence on earmarked taxes.

Primary treatment rules

- Buchanan, Lee and Wagner, Teja, McCleary, Bös, Brett and Keen, Marsiliani and Renström, Bird and Jun, Cashin et al., WHO, Prichard, and Allen and Radev are treated as core earmarking sources.
- Flores-Macías, Luttmer and Singhal, Blanchet et al., Stivender et al., Jones, Moore et al., Wilkinson, and Gwilliam and Shalizi are used for particular mechanisms or outcomes.
- Ahmad and Brosio and Smart and Bird are explicitly identified as intergovernmental-transfer evidence, not direct evidence on national earmarked taxes.

Supporting treatment rules

- Das et al. are used for coding architecture and an adjacent implementation mechanism, not as an earmarking study.
- OECD/AUC/ATAF is used only for the measurement point that standard tax classifications omit destination; its contextual revenue statistics are excluded.
- Mankiw et al. supply only a general-fund optimal-tax benchmark and are not used as evidence on earmarking.
- Akitoby et al. are omitted from the review body because their general DRM material belongs in the separately drafted introduction.
- Chelsky is used only to document tax-policy reversal outside earmarking; the source does not show that earmarking prevents reversal.

2. The object of study, boundary cases, and the legitimacy of de jure coding

Author's methodological finding: The legal revenue assignment is a valid first object because it defines the treatment and risk set, but it must be separated from transfer, additionality, expenditure quality, and behavioral outcomes.

2.1 Operative definition

An earmark is a legally specified revenue-to-purpose assignment; exact revenue–cost balance and extrabudgetary status are not definitional.

Load-bearing evidence

- **Allen and Radev (2006, pp. 12–13):** Fiscal earmarking assigns revenue from designated sources to designated expenditures; the earmarked transaction may finance either an extrabudgetary entity or an activity inside the legislative budget.
- **Bös (2000, pp. 5–7):** A practical earmark can satisfy $R \leq C$, with general revenue covering the remaining program cost; exact equality between earmarked revenue and cost is a knife-edge case rather than a necessary definitional condition.
- **Bird and Jun (2005, Table 1 and pp. ~3–6):** Earmarks differ along at least three dimensions: expenditure specificity, tightness of the revenue–expenditure link, and presence or absence of a benefit rationale.

Supporting evidence

- **McCleary (1991, pp. 82–83, Table 1):** A specific tax may finance a specific use, a broad use, or an account that combines allocation and redistribution; close payer–beneficiary overlap distinguishes “strong” from “weak” earmarking.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 9–11, Table 1):** A common description requires revenue characteristics, expenditure characteristics, the strength of the revenue–expenditure linkage, and reported results.
- **Wilkinson (1994):** Strong versus weak and narrow versus broad earmarking distinguish a binding benefit-linked arrangement from a largely formal assignment whose tax and expenditure can move independently.
- An earmark is treated as a revenue-assignment and budget-architecture device, not as a fiscal rule; it sets no numerical ceiling, floor, or target for the deficit, debt, total expenditure, or total revenue.

2.2 User charges, social contributions, grants, and budget location

User charges, social contributions, conditional grants, and budget location are adjacent dimensions rather than interchangeable forms of earmarked taxation.

Load-bearing evidence

- **Lee and Wagner (~1991, pp. 110, 112–113):** A user charge directly prices a service, while an earmarked tax is a second-best quasi-price that taxes a good correlated with use of the financed service.
- **Allen and Radev (2006, pp. 8–9):** An earmark can be managed by the finance ministry, a line ministry, or an autonomous entity, and each of these arrangements may be on-budget or extrabudgetary.
- **Smart and Bird (2009, pp. ~4–5):** An earmarked grant is a transfer with a spending condition and is not an earmarked tax; fungibility and accountability mechanisms may be analogous, but need targeting and bailout insurance are specific to intergovernmental finance.

Supporting evidence

- **Teja (1988, pp. 9–10):** An earmarked tax may substitute for a user charge where an impure public good is excludable but direct charging is administratively impractical; a general-fund subsidy may still be required for external benefits.
- **Allen and Radev (2006, pp. 13–14):** Social-security and some public-health funds are the clearest benefit-principle cases because contributions are linked to covered groups, but contributions can become de facto general revenue when the fund is integrated into a pay-as-you-go budget.
- **Ahmad and Brosio (2008, pp. ~3–7):** Revenue sharing and conditional transfers can reduce subnational autonomy and effort; that conclusion applies to transfer-dependent subnational finance and is not a universal result about national tax hypothecation.
- **Allen and Radev (2006, p. 9):** Earmarking and extrabudgetary status are not coterminous; many earmarked funds operate inside the budget, and many EBFs receive transfers, charges, borrowing, asset proceeds, or donor funds rather than earmarked taxes.

2.3 Resource-revenue funds as a boundary case

Resource-revenue funds enter the core object only when a tax or compulsory levy is legally dedicated to a specified use; royalty-, asset-income-, saving-, or stabilization-based funds remain separately coded boundary cases.

Load-bearing evidence

- **Allen and Radev (2006, p. 8):** Stabilization funds, savings funds, and financing funds are distinct EBF objectives commonly associated with oil and other nonrenewable-resource revenue.
- **Allen and Radev (2006, p. 5 n. 4):** The Norwegian State Petroleum Fund illustrates that a resource fund can be fully integrated with the annual budget, lack an independent management board, and remain subject to parliamentary and finance-ministry decisions.
- **Allen and Radev (2006, p. 20):** Oil-revenue funds require separate long-term presentation and analysis even when their transactions are included in a consolidated budget.
- **Ahmad and Brosio (2008):** Natural-resource revenue sharing raises separate problems of geographic concentration, volatility, intergovernmental derivation formulas, and horizontal inequality.

Supporting evidence

- **Allen and Radev (2006, p. 17):** The paper reports an IMF finding that oil-country saving performance did not appear to depend on the existence of an oil fund; Allen and Radev present this as evidence that formal fund creation is not sufficient.
- A petroleum fund financed by a royalty, production share, or asset return and used for stabilization or saving is recorded as an adjacent resource-fund case rather than automatically counted as an earmarked tax.
- A defined mining, petroleum, or windfall tax legally dedicated to a specified expenditure or fund falls within the tax-earmarking inventory, with a resource-revenue flag and separate outcome analysis.

2.4 Why de jure coding is a legitimate first object

De jure coding identifies the formal commitment, governance, and implementation architecture even though it cannot establish whether the rule was executed, additional, salient, or behaviorally effective.

Load-bearing evidence

- **Allen and Radev (2006, pp. 20–21):** Legal provisions can specify budget preparation, accounting, reporting, borrowing, audit, legislative scrutiny, revenue collection, and management accountability; statutory architecture therefore contains operational attributes beyond the policy label.
- **Allen and Radev (2006, pp. 24–25):** A legal inventory can establish whether an EBF has an adequate basis for financial management and reporting, whether its budget is presented to the legislature, and whether it is subject to external audit.
- **Bird and Jun (2005, Table 1):** De jure information can classify the source, destination, formal tightness, benefit rationale, and contractual character of an earmark even when its marginal spending effect remains unknown.
- **Bird and Jun (2005, pp. ~5–6):** The distinction between symbolic and substantive earmarking makes legal coding necessary rather than pointless: symbolic earmarks must first be identified before their lack of marginal bite can be measured.
- **Prichard (2015, pp. 97, 102):** Ghana's Communications Tax was politically presented as financing youth employment, but the statute assigned only 20 percent, contained no dedicated budget line or oversight committee, and left the public largely unaware of the discrepancy; de jure coding distinguishes the enacted rule from the broader political claim.

Supporting evidence

- **Allen and Radev (2006, p. 7):** GFSM classification emphasizes economic characteristics rather than legal form; this limits what legal coding can say about sector classification but does not eliminate the separate question of what obligation the law created.
- **WHO (2016, p. ~8 and Table 2):** Poland enacted a 0.5 percent tobacco-excise earmark but never adopted the regulation needed to channel funds to health; a layered legal inventory can record the statute, the missing implementing instrument, and the resulting zero transfer.
- **Cashin, Sparkes, and Bloom (2017, ~p. 17):** Botswana's tobacco funds remained unused for more than a year because no channeling mechanism existed; recording transfer machinery separately from the dedication identifies this failure mode.
- **Kimwell et al. (2022, ~pp. 32–33):** The Philippine soft earmark operated as budgetary attribution rather than traceable fund flow; statutory designation, annual appropriation, and actual expenditure therefore require separate variables.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 9–11):** Their typology records revenue characteristics, expenditure characteristics, linkage, flexibility, time horizon, accountability, and reported results; their six-country table records policy, revenue source, expenditure purpose, implementation, and results.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 7–11):** Their literature-assembled JLN database covers more than 80 countries but is self-updating, uneven in evidence quality, and not a census; it is a starting inventory rather than an execution database.
- **Das et al. (2026, ~pp. 10, 16–31):** A fixed prompt, explicit coding rubric, retained source excerpt, human benchmark, predictive validation, and comparison with an independent series provide an auditable model for scaling legal and narrative coding; the paper does not itself code earmarks.
- **OECD/AUC/ATAF (2025, §29, ~p. 292):** Standard revenue classification assigns receipts by tax base and explicitly disregards earmarking, so conventional revenue series cannot recover legal dedication.
- No IMF source in the supplied corpus states the OECD/AUC/ATAF tax-classification point in equivalent terms; Allen and Radev address consolidation and institutional classification of EBFs rather than whether tax tables preserve destination labels.
- The de jure record should contain the enabling instrument, tax or levy, rate or share, destination, effective date, sunset or review clause, amendment authority, on/off-budget status, transfer rule, appropriation discretion, governing body, reporting, audit, and collection arrangements.
- De facto variables should separately record collection, transfer, release, expenditure, offsetting changes in general-fund support, balances, borrowing, diversion, suspension, amendment, repeal, public awareness, and service outputs.

2.5 Feasibility of Bird and Jun's marginal-funding-source test

Bird and Jun's marginal-funding-source test cannot be implemented from statutes alone, but de jure data can identify the cases in which an execution-based test is feasible.

Load-bearing evidence

- **Bird and Jun (2005, pp. ~5–6):** Their substantive test asks whether earmarked revenue is the marginal source that changes spending, not whether it finances a positive average share of the program.
- **Bird and Jun (2005, pp. ~5–6):** The authors describe identification of the genuinely incremental source as a difficult empirical exercise.
- **Bird and Jun (2005, Tables 5–6):** Korea's transportation accounts illustrate the required evidence: revenue composition over time and identification of the general-account transfer as the swing source.
- **Bird and Jun (2005):** A statute can establish exclusivity, a fixed percentage, a special account, and formal restrictions, but cannot by itself show whether general-fund support fell when earmarked revenue rose.
- **Bird and Jun (2005):** The test is most feasible for separately reported special accounts or funds with complete revenue and expenditure statements, such as road funds, social-insurance funds, and Korea-style transportation accounts.
- **Bird and Jun (2005):** The test is least feasible for broad health or education earmarks where earmarked revenue finances only part of a large sector and general-fund allocations remain discretionary.

Supporting methodological implications

- A de jure inventory can code *potential marginality*—exclusive financing, full-cost coverage, automatic transfer, separate accounts, and restrictions on substitution—but actual marginality requires executed revenue and expenditure data over time.
- The de jure inventory is therefore a legitimate first-stage sampling and stratification device, not a substitute for the marginal-funding-source test.

3. Political authorization: passage, consent, and coalition formation

At the political-authorization gate, earmarking can increase stated support and facilitate passage, but the evidence does not equate that effect with payer consent, compliance, or welfare.

3.1 Theoretical predictions

Theory predicts authorization gains through disaggregated choice, quasi-pricing, signaling, and commitment, while also allowing symbolic or discriminatory political support.

Load-bearing evidence

- **Buchanan (1963, p. 459):** General-fund financing is modeled as a tie-in sale that forces citizens to purchase a bundle of services; earmarking permits service-by-service adjustment against the associated tax price.
- **Buchanan (1963, pp. 461–463):** Earmarking and general-fund financing produce the same choice only at the full-equilibrium budget ratio; a non-equilibrium bundle distorts the mix selected by the decisive voter.
- **Buchanan (1963, pp. 465–467):** Groups supporting different services may prefer different financing regimes because general-fund bundling redistributes taxpayer surplus among them.
- **Lee and Wagner (~1991, pp. 112–116):** A complementary tax–service pairing can operate as a quasi-price, while item-by-item tax-and-spend decisions can make voter choice more determinate than a general-fund package.
- **Lee and Wagner (~1991, pp. 115–116):** In the Brennan–Buchanan mechanism, a politician dependent on a complementary tax can raise more revenue only by supplying a service that sustains demand for the taxed good.
- **Brett and Keen (2000, pp. 10–13):** A policy-aligned but politically weak incumbent may earmark to protect revenue from a future self-interested successor and thereby sustain a higher environmental tax.
- **Brett and Keen (2000, Proposition 2, pp. 11–12):** Earmarking is chosen when its per-dollar efficiency loss is smaller than the expected loss from successor expropriation.

Supporting evidence

- **Teja (1988, pp. 2–5):** Separate earmarked tax shares can accommodate heterogeneous preferences in the Lindahl framework; the result depends on truthful preference revelation.
- **Teja (1988, pp. 5–8):** Browning-style collective-choice models allow earmarking to avoid mismatches between the median voter on budget size and the median voter on composition; Teja states that the model predicts political support but does not establish a welfare ranking.
- **Wagner (1991, pp. 3–4, 7–11):** The relevant benefit-principle test is consent by those who pay, not general popularity among people who can impose the tax on others.
- **Allen and Radev (2006, pp. 11–12):** A health or environmental fund can make a new tax more politically acceptable than an equivalent increase in general revenue, even when the macrofiscal effect is the same.
- **Allen and Radev (2006, p. 12):** A dedicated fund may also signal social recognition that a priority is underfunded without necessarily increasing its financing.
- **Moore, Prichard, and Fjeldstad (2018):** Selective earmarking is presented as one possible fiscal-contract device in low-trust settings, conditional on high transparency and credible monitoring.
- **Bös (2000, p. 22):** For fully rational taxpayers, merely making the tax–spending connection more understandable is not a distinct earmarking advantage because taxpayers can anticipate the general-budget connection.
- **Wilkinson (1994):** Transparency and voter understanding are more plausible for strong, narrow earmarks than for weak earmarks whose tax and spending decisions remain effectively independent.

3.2 Evidence on passage and stated support

Observed evidence is strongest for stated support and political passage in selected cases, not for an independent cross-country effect of earmarking.

Load-bearing evidence

- **Flores-Macías (2018, pp. 18–20):** Random assignment of an “exclusive use for public safety” statement increased stated support for a hypothetical tax by roughly 7–8 percentage points relative to the baseline treatment.
- **Flores-Macías (2018, Table 1 and Figs. 3–5):** The earmark effect was substantially larger among low-trust respondents; the earmark-by-trust interaction was reported at ($p=.043$), while income heterogeneity was weaker.
- **Flores-Macías (2018):** The treatment changed wording only; it did not create a law, accounting mechanism, executed dedication, payment obligation, or observed revenue.

- **Prichard (2015, pp. 92–93):** Ghana reintroduced VAT in 1998, implemented it at 10 percent in early 1999, and attached a subsequent 2.5-point increase to the GET Fund as a politically motivated means of securing passage.
- **Prichard (2015, p. 96):** Ghana's NHIL was a separately branded 2.5-point VAT increase earmarked to the NHIS; the government avoided calling it VAT because of the earlier political toxicity of that label.
- **Prichard (2015, pp. 101–102):** Prichard concludes that the GET Fund and NHIS earmarks were political strategies to reduce opposition and that the dedicated oversight bodies would not have emerged without resistance to additional taxation.
- **Cashin, Sparkes, and Bloom (2017, ~p. 23):** In all six focus countries, earmarks were adopted when health and finance objectives aligned; the Ghana VAT and Philippine tobacco reforms are presented as cases where a popular health purpose facilitated a tax measure.
- **Cashin, Sparkes, and Bloom (2017):** The six cases are purposively selected and do not identify the independent effect of the earmark on enactment.

Supporting evidence

- **Prichard (2015, pp. 97, 102):** The Communications Tax case shows that political packaging could exceed statutory content: only 20 percent was legally earmarked, while the public largely remained unaware.
- **Prichard (2009, pp. 25–27):** The Ghana process evidence similarly treats earmarking as an opportunistic or pre-emptive political strategy attached to already desired revenue measures.
- **Prichard (2015, pp. 59–60):** The tax-bargaining framework allows concessions to precede the tax increase; earmarking, passage, and anticipated resistance may therefore be jointly determined.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 17, 21–23):** Their six focus countries are Estonia, Ghana, Indonesia, the Philippines, South Africa, and Viet Nam.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 9–11):** Their database and typology contain revenue characteristics, expenditure characteristics, revenue–expenditure linkage, and reported results; the focus-country table adds policy, revenue source, expenditure purpose, implementation, and results.
- **Kaiser, Bredenkamp, and Iglesias (2016, ~pp. 50–70):** The Philippine health earmark formed one component of a broader sin-tax coalition involving presidential leadership, legislative bargaining, technical analysis, health advocacy, and international support.
- **Kaiser, Bredenkamp, and Iglesias (2016):** The process-tracing evidence does not identify earmarking as the sole or sufficient cause of passage.
- **WHO (2016, §§3.1–3.3):** The nine country cases describe health framing, finance–health alliances, evidence use, and opposition by tobacco interests, but contain no comparison of otherwise similar reforms with and without earmarking.
- **Stivender et al. (2016):** A staggered difference-in-differences analysis of US lotteries associates education earmarking with higher gross lottery sales in the baseline specification; the estimate becomes smaller and statistically weaker with flexible state trends.
- **Stivender et al. (2016):** Lottery purchases are voluntary and the study does not observe beliefs about dedication or the resulting education expenditure, limiting transfer to compulsory taxes.

3.3 Scope conditions on political authorization

Authorization gains hold only under paper-specific conditions concerning consent, credibility, incidence, participation, and coalition structure.

Load-bearing evidence

- **Prichard (2015, Table 2.1, ~pp. 70–71):** Tax bargaining depends on revenue pressure, scope for resistance, collective-action capacity, bargaining institutions, and political salience.
- **Prichard (2015, pp. 92–102):** Ghana's passage mechanism was mediated by parties, organized groups, elites, and electoral competition rather than demonstrated individual-level reciprocity.
- **Flores-Macías (2018):** The identified outcome is stated proposal support in a Mexican public-safety scenario; the paper does not establish actual voting, payment, compliance, or cross-sector external validity.
- **Cashin, Sparkes, and Bloom (2017, ~p. 23):** Political facilitation appears where the financed purpose is broadly popular and finance and health objectives align.
- **Kaiser, Bredenkamp, and Iglesias (2016):** The Philippine case bundles earmarking with rate restructuring, administrative reform, coverage expansion, coalition formation, and an already salient health program.

Supporting evidence

- **Buchanan (1963, p. 459):** The item-by-item result assumes a decisive median voter, single-peaked preferences, fixed individual tax prices, and negligible decision costs.
- **Teja (1988, pp. 4–8):** Preference-revelation benefits require heterogeneous preferences and truthful revelation; free riding and fixed tax shares weaken the normative result.
- **Wagner (1991, pp. 3–4):** Aggregate majority support does not establish consent by the taxed population because a majority may approve a charge imposed on a minority.
- **Wagner (1991, pp. 10–11):** The California cigarette referendum is only suggestive of majority-on-minority taxation because the paper lacks smoker-specific voting data.
- **Brett and Keen (2000, p. 20):** The political-signalling effect requires voters to believe that earmarked funds will be used as promised.
- **Bös (2000, p. 22):** A perception-only mechanism requires some departure from the paper's rational-anticipation benchmark.
- **Wagner (1991, preface p. xv):** The volume acknowledges Tobacco Institute and Bradley Foundation funding; the consent and constitutional arguments can be separated from its applied hostility to cigarette-tax earmarking.
- **Lee and Wagner (~1991):** The chapter year is inferred from the 1991 volume rather than printed in the supplied chapter.
[VERIFY year against the volume front matter before final citation.]

3.3.1 Lee and Wagner's constitutional non-discrimination premise

Lee and Wagner's Section 4 makes constitutional non-discrimination the premise that carries the pro-earmarking result.

Load-bearing evidence

- **Lee and Wagner (~1991, pp. 122–123):** Browning-style pro-earmarking results are developed in a "general taxation–general benefit" setting in which taxation is non-discriminatory and public services are positively valued by the general population.
- **Lee and Wagner (~1991, pp. 117, 122–123):** A normative demonstration that earmarking can improve a specified institutional arrangement does not show that actual earmarks emerging from ordinary majoritarian politics realize those benefits.
- **Lee and Wagner (~1991, pp. 118–120):** When a majority can impose a special tax on a disfavored group and direct the revenue to a favored program, item-by-item linkage may facilitate fiscal discrimination rather than reveal payer demand.

Supporting evidence

- **Lee and Wagner (~1991, pp. 119–122):** Their cigarette-tax example allows beneficiaries to value the penalty imposed on smokers separately from the revenue raised, creating political demand for a rate above the efficient or revenue-maximizing level.
- **Lee and Wagner (~1991, p. 118):** An efficiency-justified earmark requires a positive association between consumption of the taxed and financed goods, uniformity of that relation across people, and agency revenue that responds to efficient service provision rather than hostility toward the taxed group.
- **Lee and Wagner (~1991, p. 123):** A positive taxed-good/financed-good correlation is necessary but not sufficient.
- **Lee and Wagner (~1991, p. 123):** Comparing an ideal Wicksellian constitutional regime with rent-seeking general-fund politics does not imply that a piecemeal earmark adopted inside the same rent-seeking process improves welfare.
- **Lee and Wagner (~1991, p. 123):** Constitutional rules capable of preventing fiscal discrimination carry the load of converting the theoretical pro-earmarking mechanism into a positive institutional prediction.
- **Lee and Wagner (~1991, p. 123):** Without those constraints, the authors state that little can be said in general about the relative merits of earmarking and general-fund finance.

4. Revenue realization: compliance, net yield, tax-base effects, and durability

At the revenue-realization gate, both theory and evidence leave the sign of compliance and net yield unresolved, and the comparative durability of earmarked taxes is not established.

4.1 Theoretical predictions on compliance and yield

Theoretical mechanisms allow earmarking to raise willingness to pay while tax-base distortion, substitution, and political extraction can offset or reverse the gain.

Load-bearing evidence

- **Teja (1988, p. 12):** In developing countries with weak democratic institutions and high distrust, assurance that revenue will finance valued local services may improve compliance and substitute partly for absent representative power.
- **Teja (1988, p. 12):** The compliance proposition is an argument rather than an empirical result.
- **Luttmer and Singhal (2014, pp. 155–163):** Tax morale operates through intrinsic motivation, reciprocity, peer effects, culture, and information or behavioral imperfections.
- **Luttmer and Singhal (2014, pp. 158–159, 165):** Perceived unfairness can produce persistent compliance effects and a low-compliance equilibrium in which weak revenue, poor services, and weak morale reinforce one another.
- **Luttmer and Singhal (2014, pp. 156–160):** Strong prior beliefs about corruption or expropriation may make brief revenue-use messages too weak to update taxpayers.
- **McCleary (1991, pp. 84–86):** Linking tax and spending may overcome resistance and generate revenue that would otherwise not be forthcoming.
- **McCleary (1991, pp. 94–96):** Special levies can also cannibalize central revenue through reduced taxable activity, reduced profits, contradictory taxes and subsidies, or diversion of existing bases.

Supporting evidence

- **Bös (2000, p. 22):** A rational taxpayer should anticipate that general taxes finance public expenditure, so understanding the tax–spending connection is not by itself a special compliance argument for earmarking.
- **Bird and Jun (2005, §2.2.3):** Compliance may rise when taxpayers value the destination, but greater certainty about how revenue is spent may also reduce compliance among risk-averse free riders; the net sign is not obvious.
- **Lee and Wagner (~1991, pp. 113, 123):** A narrow earmarked excise can impose greater excess burden than a broad general tax raising the same amount.
- **Lee and Wagner (~1991, pp. 121–122):** Political demand to punish the taxed group can move the rate beyond the revenue-maximizing point and reduce the revenue available to the earmarked program.
- **Brett and Keen (2000, Proposition 2, pp. 11–12):** Credible earmarking can support an environmental tax above the no-earmarking rate but below the Pigovian rate.
- **Marsiliani and Renström (2000, pp. C125–C133):** A binding earmark can partially mitigate a time-consistency problem in energy taxation but cannot reproduce the committed optimum because the government adjusts other instruments.
- **Buchanan (1963, pp. 463–465):** The model identifies conditional effects on total public outlay, not an identified effect on tax collection or DRM.

4.2 Evidence on compliance and net revenue

The empirical corpus does not identify an SSA compliance effect and does not isolate earmarking within bundled revenue reforms.

Load-bearing evidence

- **Luttmer and Singhal (2014, pp. 156–160):** Moral-suasion and public-good-use letters in Austria, Minnesota, Argentina, and Israel generally had null effects on actual compliance, with one study reporting suggestive backfiring.
- **Luttmer and Singhal (2014, pp. 161–164):** Reminders, simplification, salience, and descriptive norms more consistently improved timely payment, pointing to attention rather than deep reciprocity as the movable margin.
- **Flores-Macías (2018):** The randomized earmark wording changed stated support but observed no tax liability, payment, enforcement, or revenue.
- **Prichard (2009, pp. 24–25, 34):** Political goodwill and organized taxpayer relations affected the government’s feasible tax take and some taxpayers’ willingness to comply; the paper does not attribute that behavior to an earmark.
- **Prichard (2015, pp. 104–106):** Ghana’s associational and flat-rate tax bargains link compliance to reduced harassment, simpler administration, and broader base expansion rather than to revenue dedication.

Supporting evidence

- **Flores-Macías and Sánchez-Talanquer (forthcoming, pp. 19–31):** National insecurity perceptions were associated with lower stated willingness to pay for a promised crime reduction; the cross-sectional study did not test earmarking or identify the proposed state-failure mechanism.
- **Stivender et al. (2016):** Education earmarking is associated with higher gross lottery sales in the baseline model; only 12 policy changes and sensitivity to state trends weaken causal confidence.
- **Jones (2013):** Education lotteries increased non-tax revenue and total revenue modestly but did not significantly increase education spending; non-education spending rose and private education donations fell.

- **Cashin, Sparkes, and Bloom (2017, ~pp. 36–37):** Philippine tobacco and alcohol excise revenue rose from approximately 0.5 percent of GDP before reform to 1.1 percent by 2015.
- **Cashin, Sparkes, and Bloom (2017):** The Philippine increase combined rate reform, tax administration, political coalition, health earmarking, and other policy changes, so the contribution of dedication is not separately identified.
- **McCleary (1991, pp. 94–96):** Turkey's earmarked funds expanded while total government revenue relative to GNP fell; the case is descriptive but demonstrates that fund-level receipts need not represent aggregate revenue mobilization.
- **Prichard (2015, pp. 92–101):** Ghana's earmarks were attached to planned revenue increases to secure passage; the evidence does not isolate an effect on post-enactment collection efficiency.

4.3 Durability and reversal

Commitment theory supplies a durability mechanism, but the empirical corpus does not establish that earmarked taxes face a lower risk of reversal.

4.3.1 Theoretical predictions

Durability follows in theory only when the dedication is credibly binding and cannot be cheaply amended or circumvented.

Load-bearing evidence

- **Brett and Keen (2000, pp. 11–13):** Earmarking protects tax revenue from expropriation by a future officeholder when the incumbent faces political uncertainty.
- **Brett and Keen (2000, pp. 20–22):** If successors can break the pledge at low cost, voters may disregard it and earmarking can reduce welfare.
- **Marsiliani and Renström (2000, p. C125):** An earmark has commitment value only when it is constitutionally or otherwise credibly difficult to change.
- **Marsiliani and Renström (2000, Proposition 3, pp. C132–C133):** A single earmarking rule cannot reproduce full commitment because a multi-instrument government can adjust other taxes and expenditures.
- **Teja (1988, pp. 11–12):** An earmarked fund can sustain intertemporal bargains where current legislators cannot otherwise bind successors.

Supporting evidence

- **Bird and Jun (2005):** Contractual or quasi-contractual commitment is a distinct rationale from benefit taxation and marginal expenditure effects.
- **Bös (2000):** Institutional sequencing and the separation of tax and spending authorities condition whether the dedication remains binding after uncertainty is resolved.

4.3.2 Evidence and unresolved comparison

Observed repeal, termination, and revision cases document reversibility without comparing earmarked and non-earmarked taxes.

Load-bearing evidence

- **Prichard (2015, p. 96):** Ghana's National Reconstruction Levy was introduced in 2001, renewed in 2004, and removed in 2007 despite an earmarked venture-capital promise attached to its extension.
- **WHO (2016, p. ~8):** Poland's inoperative tobacco earmark was formally terminated in 2015 when tobacco control moved into a general-budget program.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 23–27):** Periodic review revised a minority of South African earmarks while most were carried forward; the evidence does not compare their survival with non-earmarked taxes or programs.

Supporting evidence

- **Chelsky (2023, DRM evaluation discussion):** Tax-exemption reforms were sometimes implemented and later reversed, including reintroduction of exemptions before Pakistan's 2018 election; the source does not discuss earmarking.
- **Stivender et al. (2016):** The small number of observed lottery-earmark changes constrains estimation of the sales effect and does not provide a general durability comparison.
- The corpus does not establish whether earmarked taxes are harder to repeal, suspend, reduce, or reverse than otherwise comparable non-earmarked taxes.

- The corpus does not distinguish whether durability results from legal form, political beneficiaries, separate-fund balances, public salience, or the underlying popularity of the financed program.

4.4 Scope conditions on revenue realization

Revenue realization must be measured separately as gross receipts, net tax revenue, and net general-government revenue under explicit credibility, incidence, and tax-base conditions.

Load-bearing evidence

- **Prichard (2015, Table 2.1, ~pp. 70–71):** Visibility, incidence, media attention, remittance arrangements, collective organization, and bargaining institutions affect whether a tax generates politically meaningful resistance or negotiation.
- **Prichard (2009, pp. 32–35):** Petroleum taxes are broad and highly visible, while VAT and import duties are partly remitted by firms and may be less individually salient.
- **Luttmer and Singhal (2014, pp. 156–164):** A structural earmark may need to change deeply held beliefs rather than merely provide a short informational message.
- **Lee and Wagner (~1991, pp. 118–123):** A revenue gain tied to hostility against the taxed group is analytically distinct from consent by the taxed population.
- **McCleary (1991, pp. 94–96):** Fund-level revenue can coexist with central-budget erosion or contradictory taxes elsewhere.

Supporting evidence

- **Flores-Macías (2018):** Low-trust respondents were more responsive to the exclusive-use wording, but persistence after observing implementation was not tested.
- **WHO (2016):** A successful tobacco-control tax may shrink its own base over time, creating a sustainability conflict for programs dependent on the revenue.
- **Brett and Keen (2000, p. 20):** Revenue effects based on reassurance disappear when the commitment is not credible.
- Gross earmarked receipts, net tax revenue, net general-government revenue, and revenue relative to the counterfactual non-earmarked reform should be reported separately.

5. Budget translation: transfer, execution, additionality, and fungibility

At the budget-translation gate, legal dedication frequently fails to become an executed and additional expenditure because transfer machinery, appropriation discretion, and fungibility intervene.

5.1 Theoretical and operational predictions

A collected earmarked tax can fail at transfer, release, or marginality even when the statute is operative.

Load-bearing evidence

- **Bird and Jun (2005, pp. ~5–6):** An earmark is substantive only when the dedicated revenue changes marginal expenditure; financing a positive average share is insufficient.
- **McCleary (1991, p. 87):** A successful earmark must combine efficient pricing with automatic financing independent of recurrent general-budget decisions.
- **McCleary (1991, pp. 88–90):** Increasing returns, indivisibilities, externalities, and general revenue needs usually require fund deficits or surpluses, re-establishing links with the general budget and undermining automaticity.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 12–17):** Fungibility allows a dedicated inflow to release general revenue, so the target program may receive little or no net increase.

Supporting evidence

- **Teja (1988, pp. 9–10):** Joint financing can be economically appropriate where earmarked payments cover the private-benefit portion and general revenue covers external benefits.
- **Bös (2000, pp. 5–7):** An earmark may legitimately under-cover program cost; general-fund top-up is compatible with the practical definition.
- **Bös (2000, pp. 14–16):** Sequential taxation under uncertainty can generate involuntary surpluses because revenue is set to cover the highest-cost state.

- **Smart and Bird (2009, pp. ~4–10):** Conditional grants may be fungible where recipients have enough own-source revenue to offset them; this is adjacent evidence on transfers rather than national earmarked taxes.
- **Allen and Radev (2006, pp. 8–9):** The same legal dedication can operate inside the budget, through a separate account, or through an autonomous EBF, with different execution and control arrangements.

5.2 Evidence on transfer and release

Operational evidence documents missing channeling rules, frozen or diverted funds, and uneven execution after enactment.

Load-bearing evidence

- **WHO (2016, p. ~8):** Poland's 0.5 percent statutory tobacco earmark generated no health transfer because the required channeling regulation was absent.
- **WHO (2016, pp. ~6–7):** India's tobacco cess entered the national exchequer, required the health ministry to apply for funds, and allowed unclaimed resources to finance other purposes.
- **WHO (2016, p. ~28):** Egypt's student-health-insurance earmark did not cover program services and required general-revenue supplementation.
- **Cashin, Sparkes, and Bloom (2017, ~p. 17):** Botswana's earmarked tobacco funds remained unused for more than a year because no transfer channel had been established.
- **Cashin, Sparkes, and Bloom (2017, ~p. 17):** The Philippine Department of Budget and Management retained discretion over release despite strong statutory language.
- **McCleary (1991, pp. 90–92):** Road-fund resources were frozen in Ghana, diverted in Mali, and weakened by nonpayment of fuel taxes by public enterprises in Zaire.
- **McCleary (1991, pp. 90–92):** The Central African Republic, Colombia, Mali, and Zaire obtained at most temporary procedural benefits from their road funds; Ghana showed some success but also experienced freezing.

Supporting evidence

- **Kaiser, Bredenkamp, and Iglesias (2016):** Philippine program execution varied substantially across components, with high execution for insurance transfers and materially lower execution for facilities and some other programs.
- **Kaiser, Bredenkamp, and Iglesias (2016, reported execution figures):** The summary reports approximately 100 percent execution for the national health-insurance component, 57 percent for the Health Facilities Enhancement Program, 78 percent for Millennium Development Goal programs, and 81 percent for medical assistance.
- **Kimwell et al. (2022, ~pp. 32–33):** Soft earmarking and annual attribution prevented clean tracing of particular tax receipts into particular expenditures.
- **Kimwell et al. (2022):** Median disbursement for the Philippine health-facilities program was reported at approximately 74 percent over 2014–21, with procurement and local capacity constraints.
- **Das et al. (2026, Tables 15–17, ~pp. 64–66; Appendix D, ~pp. 82–83):** A "favorable political environment" combines legislative cohesion, no current or next-quarter national election, and absence of material unrest, gridlock, or policy paralysis.
- **Das et al. (2026):** Stronger realized-spending pass-through in favorable environments is consistent with lower legislative and implementation frictions; it is not a finding about public support, earmarking, or a directly estimated mediation mechanism.
- **Das et al. (2026):** The coded actions mix announcements, legislation, and implementation, so the estimates do not provide a clean de jure-to-execution decomposition.

5.3 Evidence on additionality and expenditure substitution

Evidence on additionality is mixed and often shows substitution across general funds, other sectors, or private finance.

Load-bearing evidence

- **Bird and Jun (2005, \$3.5):** Deran found no relationship between earmarking and expenditure levels in US states.
- **Bird and Jun (2005, \$3.5):** Eklund reported higher highway spending in developing-country cross-sections with earmarking.
- **Bird and Jun (2005, \$3.5):** Dye and McGuire found that earmarked highway taxes increased highway spending by only a fraction of the dedicated revenue because general-fund support fell.
- **Bird and Jun (2005, \$3.5):** Dye and McGuire found little effect of education earmarking on education spending.
- **Bird and Jun (2005, \$3.5):** Borg and Mason found that lottery earmarking for schools reduced school aid because general funds fell by more than the earmarked receipts.
- **Bird and Jun (2005, Korea tables):** Korea's general-account transfers appeared to be the marginal source for transportation and likely education despite extensive nominal earmarking.

- **Prichard (2015, pp. 102–103):** In the early Ghana period, total health and education spending rose by more than the new earmarked revenue, so aggregate additionality was not immediately displaced.
- **Prichard (2015, pp. 102–103):** Non-wage spending in core health and education budgets fell, indicating fungibility in expenditure composition even where headline sector spending increased.
- **Prichard (2015, pp. 102–103):** Officials expected later substitution outside the NHIS, making the early additionality result time-bounded.
- **Cashin, Sparkes, and Bloom (2017, ~p. 16):** A Japan–World Bank 11-country UHC study found no strong relationship among earmarking, health-revenue generation, and UHC achievement.
- **Cashin, Sparkes, and Bloom (2017, ~p. 16):** Brazil’s constitutional health-expenditure earmarks did not produce the intended increase at all government levels.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 24–27, 32–33):** Ghana’s health share of total government spending returned to its pre-earmark level over the longer period, despite the earmark’s importance to the NHIS.

Supporting evidence

- **Teja (1988, Colombia section):** Tied health and education transfers were used partly to replace locally financed spending; Teja presents this as evidence that legal restriction need not produce incremental expenditure.
- **Jones (2013):** US education lotteries increased non-tax and total revenue but did not significantly increase education spending.
- **Jones (2013):** Non-education spending rose and private education donations fell, demonstrating public and private substitution around a labeled education revenue source.
- **Ahmad and Brosio (2008):** Bolivia’s ring-fenced HIPC municipal accounts accumulated balances and were not followed by clear social-outcome gains; the finding is drawn from cited prior work rather than an original causal analysis.
- **Smart and Bird (2009, Box 1):** Colombian transfer conditions bound closely in most small municipalities but were offset more freely by large urban governments with greater own-source revenue.
- **Smart and Bird (2009):** The Colombia result concerns intergovernmental grants and supports a capacity-dependent fungibility hypothesis rather than a direct national-tax estimate.
- **Kimwell et al. (2022):** Time-series specifications associate the Philippine earmarking period with higher health appropriations but also identify an inverse relation between sin-tax proceeds and the government counterpart, consistent with offsetting.
- **Kimwell et al. (2022):** The small time-series sample and bundled reform prevent a clean causal additionality claim.

5.4 Scope conditions on budget translation

Translation into incremental spending is most plausible for narrow, executed, and binding links, and least plausible under broad sectoral assignments, fiscal stress, and offsetting capacity.

Load-bearing evidence

- **Bird and Jun (2005):** An earmark financing all and only a discrete activity is more likely to bind than one financing a small share of a broad sector.
- **McCleary (1991, pp. 88–90):** Infrastructure characterized by indivisibility, increasing returns, or externalities is unlikely to sustain both efficient pricing and a self-balancing automatic fund.
- **McCleary (1991, pp. 90–92, 101):** Formal protection is most likely to be overridden when the general budget is under stress.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 23–27):** A purpose that is too broad invites fungibility, while one that is too narrow creates rigidity.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 23–27):** Release valves and periodic review can reduce rigidity but also make the arrangement resemble ordinary budgeting.

Supporting evidence

- **Smart and Bird (2009, Box 1):** Recipient own-source capacity determines whether a transfer condition can be undone by offsetting expenditure.
- **Kaiser, Bredenkamp, and Iglesias (2016):** Philippine budgets began adjusting before all revenues flowed, making anticipation and pre-trends relevant to causal evaluation.
- **Prichard (2015, pp. 59–60):** Pre-emptive bargaining means the planned tax, earmark, expected opposition, and accompanying expenditure commitments may be jointly determined.
- **Teja (1988, pp. 14–15):** Joint financing, adjustable rates or bases, and periodic review can reduce rigidity.
- **Cashin, Sparkes, and Bloom (2017):** The corpus does not establish a general superiority of soft over hard earmarks; the better-performing soft cases also retain more ordinary budget discretion.

6. Expenditure quality and fund governance

At the expenditure-performance gate, dedicated finance has no independent quality effect; governance and implementation capacity determine whether resources become efficient and equitable services.

6.1 Theoretical and operational predictions

Theory and operational guidance make purchaser incentives, effort, audit, and accountability—not earmarking alone—the mechanisms of expenditure quality.

Load-bearing evidence

- **Allen and Radev (2006, pp. 14–16):** An agency model can generate microeconomic efficiency when the entity operates as a purchaser, faces performance incentives, and is embedded in a strong governance and financial-management framework.
- **Allen and Radev (2006, pp. 15–16):** Financial autonomy without adequate controls creates risks of unauthorized asset use, excessive borrowing, corruption, weak reporting, and neglected audit.
- **Allen and Radev (2006, pp. 19–20):** An efficient road-fund-style agency should have a clear mission, physical and financial output indicators, a defined resource envelope, and a board free of producer interests.
- **Lee and Wagner (~1991, p. 118):** A defensible earmark requires agency revenue to respond to efficient service provision rather than the agency's ability to mobilize hostility toward the taxed population.
- **Teja (1988, pp. 10–11):** Earmarking can force a budget-maximizing bureaucracy to fund neglected maintenance rather than preferred new investment.
- **Teja (1988, pp. 11–12):** The same commitment technology can protect productive expenditure or be captured to protect a special interest's tax burden.

Supporting evidence

- **Bös (2000, pp. 18–20):** A vote-maximizing government can obtain exact earmarking by discouraging the spending minister's effort so that costs rise to absorb revenue; political support can therefore coexist with inefficiently high cost.
- **Gwilliam and Shalizi (1999):** Second-generation road funds seek to combine user charges, user representation, secure maintenance finance, commercial management, and audit.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 23–27):** PFM quality, absorptive capacity, release mechanisms, and periodic review condition whether the health sector can use earmarked resources.
- **WHO (2016):** Ministry-managed, autonomous, and semi-autonomous tobacco funds impose different demands on managerial independence, oversight, and accountability; the report does not estimate a causal governance ranking.
- **Smart and Bird (2009):** Earmarked grants can clarify responsibility or create fiscal illusion, credit claiming, and blame shifting; this is adjacent accountability theory rather than tax-earmark evidence.

6.2 Evidence on expenditure quality and governance

Cases show selective maintenance, purchasing, and utilization gains alongside procurement delays, capture, inequity, and weak causal attribution.

Load-bearing evidence

- **Gwilliam and Shalizi (1999):** Ghana's second-generation road-fund arrangements were associated in cited operational evidence with unit costs roughly 15–20 percent lower, attributed to more reliable funding and contracting.
- **Gwilliam and Shalizi (1999):** Earlier road funds in Sierra Leone, the Central African Republic, Tanzania, and Zambia suffered diversion, weak governance, or failure to maintain the intended user-charge relationship.
- **McCleary (1991, pp. 92–94):** Colombia's municipal valorization charge received a favorable operational assessment because project identification preceded financing, assessed payers overlapped with beneficiaries, and general-fund diversion was limited.
- **McCleary (1991, pp. 90–92):** The apparent Ghana road-fund success was associated with an independently strong government commitment to road rehabilitation, leaving the earmark effect confounded.
- **Prichard (2015, pp. 103–104):** Ghana's GET Fund and NHIS had boards and Auditor-General oversight outside standard finance-ministry mechanisms; the evidence on better oversight was described as ambiguous and under-researched.
- **Prichard (2015, p. 104):** Public pressure forced restoration of GET Fund resources previously appropriated by government.
- **Prichard (2015, p. 104):** The GET Fund also faced detailed misuse and corruption allegations; Prichard's assessment of spending-quality benefits remains qualified.

- **Kimwell et al. (2022):** Procurement delays, local implementation capacity, program proliferation, and crowding among health programs reduced the efficiency of Philippine earmarked appropriations.
- **Kimwell et al. (2022):** Family-health expenditure was associated with favorable outcomes, immunization spending with a perverse result, and tuberculosis spending with no clear effect; the authors do not claim causal program effects.
- **Kimwell et al. (2022):** Poverty targeting was weak in parts of the Philippine allocation system.

Supporting evidence

- **Allen and Radev (2006, p. 17):** Uganda met much of its road-maintenance requirement through the ordinary budget and committed more resources per kilometer than nearby “securely funded” countries with road funds.
- **Allen and Radev (2006, p. 18):** Ghana’s proliferation of statutory funds in roads, social security, and mining adversely affected overall budget management.
- **Kaiser, Bredenkamp, and Iglesias (2016):** Philippine implementation showed high execution in some transfer programs but weaker execution in infrastructure, targeting, and local-government components.
- **Kaiser, Bredenkamp, and Iglesias (2016):** The tobacco-region allocation was politically important but exposed to corruption, patronage, and delay after judicial intervention in pork-barrel arrangements.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 24–27):** Ghana’s NHIS depleted reserves and required a government bailout in 2015.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 36–37):** The Philippine revenue surge generated absorptive-capacity pressures despite substantial health-budget expansion.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 23–27):** South African reviews carried over approximately 80 percent of earmarks and revised about 20 percent, illustrating both continuity and the need to adjust obsolete allocations.
- **WHO (2016):** Thailand’s autonomous ThaiHealth, Viet Nam’s semi-autonomous Tobacco Control Fund, ministry-managed funds, and Egypt’s GAHI arrangement illustrate governance variety without identifying an autonomy effect.
- **Blanchet, Fink, and Osei-Akoto (2012):** Ghanaian NHIS coverage was associated with greater formal health-care utilization among women in Accra after matching, but enrolment remained selective by wealth and other characteristics.
- **Blanchet, Fink, and Osei-Akoto (2012):** The study evaluates an insurance program financed largely by earmarked revenue and does not estimate the effect of the earmark itself.
- **Ahmad and Brosio (2008, pp. ~6–7):** Mexican education and health earmarks were associated in the cited institutional evidence with ratcheting personnel expenditure without corresponding service-quality gains.
- **Allen and Radev (2006, pp. 18–19):** Donor-funded EBFs can create “islands” that attract scarce skilled staff and weaken the wider public administration even where the individual entity performs well.

6.3 Scope conditions on expenditure quality

Quality gains are conditional on complete accounts, purchaser discipline, independent oversight, and absorptive capacity, conditions for which causal evidence remains thin.

Load-bearing evidence

- **Allen and Radev (2006, pp. 19–22):** A dedicated agency should operate principally as a purchaser, maintain output indicators and complete accounts, face internal and external audit, and permit central access to cash information.
- **Allen and Radev (2006, p. 19):** Private-sector board representation is useful only when the board is independent of suppliers, producers, and trade-union interests.
- **Allen and Radev (2006, pp. 15–16):** A weak central budget and control system is unlikely to enforce higher standards on an autonomous subordinate fund.
- **Cashin, Sparkes, and Bloom (2017):** Revenue expansion without absorptive capacity can produce balances, delayed procurement, and inefficient program proliferation.
- **Gwilliam and Shalizi (1999):** Road-fund governance is most defensible in an intermediate institutional range where ordinary budgeting is weak enough to motivate reform but governance is strong enough to protect the fund.

Supporting evidence

- **WHO (2016):** Formal autonomy does not substitute for enabling regulation, transparent allocation, audit, and a functioning transfer mechanism.
- **Bös (2000):** Political support for exact revenue–cost matching need not imply efficient production.
- **Blanchet, Fink, and Osei-Akoto (2012):** Program utilization evidence does not establish equity, universal coverage, or the causal contribution of the financing label.
- Evidence on expenditure quality is thinner and less causally identified than evidence on political passage or accounting flows.

7. Budget comprehensiveness and extrabudgetary funds

Earmarking becomes a budget-comprehensiveness problem when a dedicated transaction or fund falls outside consolidated reporting, common controls, or unified cash and debt management; the risk is not inherent in every earmark, and earmarking is not a fiscal rule.

7.1 Earmarking and extrabudgetary status

Earmarking and extrabudgetary status are distinct, but their intersection creates the relevant exposure to incomplete budget coverage and weaker oversight.

Load-bearing evidence

- **Allen and Radev (2006, pp. 4–5):** EBFs are general-government transactions, often with separate banking or institutional arrangements, that are excluded from the annual state or subnational budget law.
- **Allen and Radev (2006, pp. 8–9):** EBFs may be financed by earmarked revenue, budget transfers, user charges, asset proceeds, borrowing, or donor funds and may be managed centrally, by line ministries, or by autonomous agencies.
- **Allen and Radev (2006, pp. 8–9):** The same activities can be organized as earmarked on-budget funds; an earmark therefore need not create an EBF, and an EBF need not be tax-financed.

Supporting evidence

- **Allen and Radev (2006, p. 6, Table 1):** In the 2002–04 GFS sample, EBF outlays excluding social security averaged 13.4 percent of central-government outlays and social-security funds 30.2 percent; the estimate is not SSA-specific and rests on small country samples.
- **Allen and Radev (2006, p. 7):** Statistical classification should follow the economic characteristics and accounts of the institutional unit rather than its legal title.
- **Allen and Radev (2006, pp. 12–13):** Fiscal earmarking is the assignment of designated revenue to designated expenditure, whether the activity is inside or outside the legislative budget.

7.2 The budget-comprehensiveness problem

The budget-comprehensiveness cost arises from incomplete fiscal information and fragmented authority rather than from the statutory revenue label alone.

Load-bearing evidence

- **Allen and Radev (2006, pp. 16–17):** Missing or delayed EBF information can distort assessment of government size, aggregate demand, investment, saving, the tax burden, social protection, fiscal sustainability, and contingent liabilities.
- **Allen and Radev (2006, pp. 17–18):** EBFs can weaken aggregate expenditure control, strategic allocation, operational and cash management, legislative scrutiny, and the ability to alter entrenched revenue assignments when priorities change.
- **Allen and Radev (2006, p. 18):** Proliferation can atomize governance and damage the aggregate budget system even where individual funds appear benign; Ghana's statutory funds in roads, social security, and mining are cited as adversely affecting overall budget management.

Supporting evidence

- **Allen and Radev (2006, pp. 17–18):** Separate borrowing and banking arrangements can obscure liabilities, increase financing costs, and weaken cash control.
- **Allen and Radev (2006, pp. 17–18):** Reduced transparency can facilitate inefficiency, misuse, rent seeking, monopoly power, and political or administrative "empires."

7.3 Scope conditions and reform

An earmarked EBF is defensible only when it is consolidated with the budget and subjected to common reporting, control, collection, audit, and cash-management requirements; otherwise integration with the ordinary budget is preferred.

Load-bearing evidence

- **Allen and Radev (2006, pp. 19–22):** A retained EBF requires an economic and governance rationale, coordination with the rest of the public sector, effective consolidation with the budget, full transparency, legislative scrutiny, accounting, internal control, and internal and external audit.
- **Allen and Radev (2006, pp. 21–22):** Collection of earmarked revenue should normally be integrated into the common tax administration to use shared registration, payment, audit, enforcement, information, and scale economies; a weak administration may require parallel modernization.
- **Allen and Radev (2006, pp. 23–24):** Fiscal tables should consolidate EBF revenue, expenditure, and financing on a gross basis, eliminate only interfund double counting, and cover EBFs through cash planning, commitment control, the treasury single account, accounting, reporting, audit, and oversight.
- **Allen and Radev (2006, pp. 25–27):** EBFs remain a second-best approach where their objectives can be achieved inside a comprehensive budget, and abolition is unlikely to endure unless the budget failures that motivated the fund are corrected.

Supporting evidence

- **Allen and Radev (2006, pp. 24–25):** Systematic review should identify funds to retain, merge, commercialize, privatize, phase out, or abolish, with sunset provisions for redundant entities.
- **Allen and Radev (2006, pp. 26–27):** Political and technical gatekeeping should prevent unjustified funds from damaging the integrity of the budget system.

7.4 Classification relative to fiscal rules

Earmarks are allocation formulas rather than fiscal rules because they set no numerical constraint on a fiscal aggregate.

Load-bearing evidence

- **Allen and Radev (2006, pp. 12–13):** Earmarking is defined by assignment of a revenue stream to an expenditure purpose.
- **Allen and Radev (2006, pp. 16–18):** The associated macrofiscal concerns are incomplete consolidation, borrowing, loss of expenditure control, and allocation; the earmark itself contains no numerical deficit, debt, expenditure, or revenue limit.

Supporting classification implications

- A percentage of VAT assigned to health or education is a transfer or allocation formula, not a fiscal rule governing an aggregate.
- A separate numerical rule applied to an earmarked fund should be coded as an additional institutional feature rather than used to reclassify earmarking itself.

8. How SSA institutional conditions shift the priors

Author's evidence assessment: The SSA material supports a demand–enforceability tension in which earmarking is sought where trust and ordinary budgeting are weak, while its claimed mechanisms require credible commitment, participation, administration, and oversight; the corpus does not establish this as a universal SSA result.

8.1 Conditions specified by theory and operational analysis

The mechanisms claimed for earmarking require institutional conditions that are separate from the political demand for visible dedication.

Load-bearing evidence

- **Lee and Wagner (~1991, pp. 118, 122–123):** The quasi-price case requires a strong and broadly uniform taxed-good/financed-good relationship and constitutional constraints against fiscal discrimination.
- **Allen and Radev (2006, pp. 14–16, 21–24):** An autonomous agency requires effective central budgeting, trained staff, information systems, audit, reporting, tax administration, and cash management.
- **Gwilliam and Shalizi (1999):** Second-generation road funds require an intermediate governance window: ordinary budgeting must be deficient, but user governance, audit, and protection against diversion must remain credible.
- **McCleary (1991, pp. 82–90):** Effective earmarking requires payer–beneficiary overlap, efficient pricing, and financing automaticity, a combination that is uncommon in large national infrastructure and social programs.

Supporting evidence

- **Buchanan (1963, pp. 467–469):** The allocative case assumes negligible decision costs, stable individual tax prices, and an effectively executed earmark; decision costs could reverse the result.
- **Teja (1988, pp. 4–8):** Preference-revelation arguments require participation, truthful revelation, and periodic reconsideration of the earmark.
- **Teja (1988, p. 12):** The compliance rationale is directed precisely at settings with high mistrust and weak representative institutions.
- **Teja (1988):** Teja also reports the objection that public-choice arguments lose relevance where meaningful citizen participation is absent.
- **Brett and Keen (2000, pp. 11–13, 20):** Low political reputation increases demand for earmarking, while actual benefits require credible enforcement of the commitment.
- **Marsiliani and Renström (2000, pp. C125–C134):** Inequality strengthens the demand for a commitment device, but only a genuinely binding rule has economic effect.
- **Cashin, Sparkes, and Bloom (2017, ~pp. 23–27):** Health earmarking requires health–finance alignment, PFM capacity, absorptive capacity, release mechanisms, and periodic review.
- **Moore, Prichard, and Fjeldstad (2018):** Selective earmarking in African fiscal bargaining requires unusually visible links and continued civil-society monitoring.
- **Flores-Macías (2018):** Low-trust respondents can be particularly responsive to a promise of exclusive use, but the experiment does not establish whether weak institutions can credibly deliver that promise.

8.2 SSA and developing-country observations

SSA cases show political adoption gains and selected operational successes alongside non-transfer, substitution, bailout, oversight, and budget-fragmentation failures.

Load-bearing evidence

- **Prichard (2015, pp. 92–102):** Ghana’s VAT-linked education and health earmarks were adopted in response to politically salient resistance and helped make revenue measures acceptable.
- **Prichard (2015, pp. 97, 102):** The Communications Tax case shows that weak parliament and weak oversight allowed the political claim to exceed the statutory dedication.
- **Prichard (2015, pp. 102–104):** Ghana’s early health and education additionality was positive at the aggregate level, while non-wage composition and governance concerns qualified the result.
- **McCleary (1991, pp. 90–92):** Four of the five reviewed road-fund countries—Central African Republic, Ghana, Mali, and Zaire—are in SSA; only Ghana received a partly favorable assessment, and even that case was driven by an independent policy commitment.
- **Cashin, Sparkes, and Bloom (2017, Ghana case):** Ghana’s earmarked NHIS funding supported the scheme’s existence but did not produce a lasting increase in health’s share of government spending and eventually required a bailout.
- **Cashin, Sparkes, and Bloom (2017, Botswana case):** Tobacco resources remained unused because the transfer mechanism was missing.
- **Cashin, Sparkes, and Bloom (2017, South Africa case):** Periodic review allowed some revision, while continued carry-over also preserved allocations that could constrain reprioritization.

Supporting evidence

- **Blanchet, Fink, and Osei-Akoto (2012):** NHIS coverage was associated with greater formal health-service use, but enrolment remained incomplete and socially selective.
- **WHO (2016, Botswana case):** A health-oriented tobacco levy could be legislated before administrative arrangements existed to spend it.
- **Gwilliam and Shalizi (1999):** Second-generation African road funds improved aspects of maintenance finance in some cases but remained vulnerable to diversion, governance weakness, and disconnection between user charges and road needs.
- **Allen and Radev (2006, p. 17):** Uganda met road-maintenance needs relatively well without a road fund, showing that ordinary budgeting can outperform formal dedication.
- **Allen and Radev (2006, p. 18):** Ghana’s proliferation of statutory funds is cited as damaging overall budget management.
- **Allen and Radev (2006, p. 22):** Developing and transition countries are less likely to possess the governance and financial infrastructure required for well-managed autonomous EBFs.
- **Moore, Prichard, and Fjeldstad (2018):** Ghanaian education and health earmarks are presented as examples of fiscal-contract bargaining rather than evidence of a general SSA revenue effect.

- **Ahmad and Brosio (2008):** Low-capacity, transfer-dependent systems can accumulate ring-fenced balances or weaken local effort; the evidence is Latin American and intergovernmental rather than SSA tax evidence.
- **Cashin, Sparkes, and Bloom (2017):** The SSA health evidence is concentrated in Ghana, South Africa, Botswana, and shorter references to other countries; the six-case sample is not representative.
- **Fonseca Corona (2024):** The broader tax-morale evidence base is dominated by quantitative cross-country work and contains few tightly identified single-country, single-mechanism studies; the review does not study earmarking.

8.3 Scope of the prior

The prior is conditional rather than uniformly negative because weak institutions can both motivate and defeat earmarking, while the direction of causality remains unresolved.

Load-bearing evidence

- **Teja (1988, pp. 14–19):** Earmarking may be a flight from poor budgeting rather than its cause, creating reverse causality in comparisons of earmarking and budget performance.
- **Bird and Jun (2005, pp. ~13–14):** The prevalence of earmarking may reflect lack of confidence in the ordinary budget process; the authors identify this reverse-causality proposition as untested.
- **McCleary (1991):** Several road and national fund systems emerged as escapes from cumbersome or dysfunctional general-budget procedures.
- **Gwilliam and Shalizi (1999):** Governance weakness can be both the reason for demanding a separate fund and the reason the fund fails.

Supporting evidence

- **Brett and Keen (2000):** Political weakness increases the value of commitment only when commitment can nevertheless be made credible.
- **Teja (1988):** High mistrust motivates the compliance argument, while absence of participation undermines the public-choice mechanism.
- **Allen and Radev (2006):** Some developing-country funds may be justified, but autonomous agency arrangements are not recommended where governance and PFM cannot sustain them.
- **Prichard (2015):** Weak oversight can allow a politically successful but fiscally thin earmark, while public monitoring can also force restoration of diverted resources.
- **McCleary (1991):** Local benefit-linked arrangements can work in developing settings even when large national fund systems perform poorly.
- The sources do not show that all SSA countries occupy the same institutional range or that governance quality is the sole moderator.
- The evidence does not identify whether institutional weakness causes earmarking failure, causes earmarking adoption, or jointly determines both.

9. What the literature does not establish

Author's evidence assessment: The corpus does not establish that earmarking raises net DRM in SSA, nor the intermediate mechanisms needed to infer that result from passage, gross receipts, or target-sector spending.

9.1 Net revenue mobilization

No supplied study identifies the causal effect of earmarking on aggregate net tax revenue in SSA.

Load-bearing evidence

- **Teja (1988):** The developing-country compliance claim is not empirically tested.
- **Cashin, Sparkes, and Bloom (2017, ~p. 27):** The authors state that the counterfactual without earmarking is unknowable in their cases.
- **Prichard (2015):** Ghanaian evidence establishes a passage and bargaining role, not an earmark-induced individual compliance effect.
- **Flores-Macías (2018):** The experiment estimates stated support rather than payment or collection.

Supporting evidence

- **Buchanan (1963):** The formal result concerns public outlay and allocation, not tax compliance or net revenue.
- **McCleary (1991):** Operational cases are selected and descriptive rather than estimates of aggregate revenue effects.
- **WHO (2016, ~p. 20):** The report disclaims a clear association between tobacco earmarking and health outcomes and does not estimate net revenue.
- **Stivender et al. (2016):** Lottery sales are a voluntary purchase and estimates are sensitive to flexible trends.
- **Jones (2013):** US lottery revenue results do not generalize directly to compulsory taxation or SSA.
- No supplied study identifies the causal effect of earmarking on aggregate tax revenue, net of tax-base erosion, administrative cost, substitution from other taxes, and reform-selection effects in SSA.

9.2 Compliance and perception

No supplied study jointly identifies legal dedication, taxpayer awareness, perceived credibility, actual compliance, and executed spending.

Load-bearing evidence

- **Luttmer and Singhal (2014):** Revenue-use messages generally fail to move observed compliance, but the studies do not test legal earmarking.
- **Flores-Macías (2018):** The treatment does not measure whether respondents believed the promise or how support changed after observing execution.
- **Prichard (2015):** Elite bargaining and protest avoidance are documented more clearly than atomistic taxpayer reciprocity.

Supporting evidence

- **Bird and Jun (2005):** The compliance sign remains theoretically ambiguous.
- **Bös (2000):** The rational-expectations benchmark excludes the bounded-rationality or salience mechanism required by a perception-only interpretation.
- No study jointly measures legal dedication, taxpayer awareness, perceived credibility, actual compliance, and executed spending.

9.3 Additionality and expenditure composition

No supplied study establishes target-sector additionality across the full sequence from legal rule to executed transfer, general-fund offset, and expenditure composition in SSA.

Load-bearing evidence

- **Bird and Jun (2005):** Average earmarked shares do not establish the marginal source of expenditure.
- **Cashin, Sparkes, and Bloom (2017):** Country accounts rarely provide a credible no-earmark counterfactual.
- **Prichard (2015):** Early Ghana additionality and later compositional substitution cannot be assigned causally to the earmark.

Supporting evidence

- **Kimwell et al. (2022):** Short time series, soft attribution, and bundled reform limit causal interpretation.
- **Kaiser, Bredenkamp, and Iglesias (2016):** Program execution differences do not identify the effect of the financing rule.
- **Jones (2013):** Education-lottery results identify substitution in the United States but not the operation of SSA tax earmarks.
- There is no multi-country SSA panel combining de jure rules, executed transfers, general-fund offsets, expenditure composition, and fund balances.

9.4 Expenditure quality and governance

No supplied study isolates the causal effect of a separate fund, board, audit rule, or purchaser arrangement on expenditure quality.

Load-bearing evidence

- **Allen and Radev (2006):** The paper supplies governance criteria but does not estimate their causal effects.
- **Gwilliam and Shalizi (1999):** Road-fund cost and maintenance claims rely substantially on operational and cited evidence.
- **Prichard (2015):** Oversight and spending-quality gains in Ghana are explicitly described as ambiguous and under-researched.

Supporting evidence

- **WHO (2016):** Ministry, autonomous, and semi-autonomous management models are compared descriptively.
- **Kimwell et al. (2022):** Program-output associations do not identify causal spending effectiveness.
- **Blanchet, Fink, and Osei-Akoto (2012):** Insurance utilization effects do not identify the financing rule's contribution.
- No study isolates whether a separate fund, board, audit rule, or purchaser arrangement improves service quality relative to the same revenue managed through the ordinary budget.

9.5 Institutional form and durability

No supplied study shows that hard or soft earmarks perform better or that earmarked taxes are more durable than comparable non-earmarked taxes.

Load-bearing evidence

- **Cashin, Sparkes, and Bloom (2017):** The corpus does not demonstrate a general soft-over-hard performance ranking.
- **Marsiliani and Renström (2000):** The formal model defines meaningful commitment as binding but does not measure real statutory tiers.
- **Brett and Keen (2000):** Commitment value is theoretical and may reverse when renegeing is possible.

Supporting evidence

- **Chelsky (2023):** The reversal evidence concerns tax exemptions rather than earmarked taxes.
- **Prichard (2015) and WHO (2016):** Individual earmarked measures were removed, terminated, or revised, but no matched non-earmarked comparison is available.
- Whether earmarked taxes are harder to reverse than non-earmarked taxes remains open.
- No study separates repeal, rate reduction, suspension, transfer interruption, diversion, and informal circumvention as distinct durability outcomes.

9.6 SSA coverage and identification

The SSA evidence is neither representative nor designed to estimate a regional average treatment effect.

Load-bearing evidence

- **Teja (1988, pp. 14–19):** Earmarking may be a flight from poor budgeting rather than its cause, creating reverse causality in comparisons of earmarking and budget performance.
- **Bird and Jun (2005, pp. ~13–14):** The prevalence of earmarking may reflect lack of confidence in the ordinary budget process; the authors identify this reverse-causality proposition as untested.
- **McCleary (1991):** Several road and national fund systems emerged as escapes from cumbersome or dysfunctional general-budget procedures.
- **McCleary (1991):** The African road-fund evidence is an operational review of a small set of cases.
- **Prichard (2009; 2015):** Ghana is a rich process-tracing case but cannot establish an SSA average effect.

Supporting evidence

- **Cashin, Sparkes, and Bloom (2017):** The 80-plus-country database is literature assembled, self-updating, and not a census.
- **WHO (2016):** The nine-country tobacco series is selected for policy lessons and advocacy rather than representativeness.
- **Das et al. (2026):** Nine SSA countries appear in the spending-action database, generally with short samples, but earmarking is absent and policy stages are mixed.
- **Fonseca Corona (2024):** The tax-morale review identifies a shortage of single-country, single-mechanism research but does not fill the earmarking gap.
- No representative inventory establishes the prevalence, statutory strength, destination, or survival of earmarked taxes across SSA.
- No supplied paper uses a quasi-experimental design based on the introduction, repeal, suspension, or legal-strength change of an SSA earmarked tax.

9.7 Boundary and welfare questions

The corpus supplies no common welfare metric or settled boundary rule that permits all tax earmarks, resource funds, social contributions, and intergovernmental transfers to be evaluated as one treatment.

Load-bearing evidence

- **Lee and Wagner (~1991):** Constitutional non-discrimination is required for their normative case, but the corpus does not operationalize it across countries.
- **Allen and Radev (2006):** Resource, social-security, road, donor, and agency funds require different analyses despite sharing an EBF label.

Supporting evidence

- **Buchanan (1963), Teja (1988), and Bös (2000):** The formal welfare results depend on incompatible or difficult-to-observe assumptions about voters, information, tax prices, timing, and political objectives.
- **McCleary (1991):** Strong local benefit-linked earmarks do not establish the desirability of broad national social-sector earmarks.
- No common welfare metric integrates political passage, excess burden, administrative cost, additionality, service quality, distribution, and macrofiscal fragmentation.
- The literature does not supply a settled rule for pooling tax-financed resource funds, royalty-financed savings funds, and intergovernmental resource-sharing arrangements in one treatment category.

10. Proposed synthesis table

Author’s measurement finding: A synthesis table must code the fiscal stage and counterfactual before direction because “positive” or “negative” changes meaning across passage, revenue, spending, and service outcomes.

10.1 Proposed columns

The minimum table must distinguish claim, evidence design, setting, outcome direction, and paper-specific scope condition, with additional fields separating legal form from behavior.

Primary columns

- Citation.
- Focal claim.
- Fiscal stage or channel.
- Evidence type: formal model; normative argument; comparative-institutional analysis; operational review; process tracing; accounting decomposition; descriptive case series; randomized survey experiment; matching-based observational study; time-series model; staggered difference-in-differences; narrative proxy-SVAR/LP-IV; classification or methods paper.
- Identification or counterfactual.
- Setting class: SSA; other LMIC; advanced economy; mixed/global; abstract.
- Unit and named countries.
- Instrument: tax, levy, social contribution, grant, EBF, resource fund, lottery receipt, or general fiscal action.

Supporting columns

- De jure/de facto layer measured.
- Outcome measured.
- Direction: positive, negative, zero/not established, mixed, or condition-dependent, always relative to the named outcome.
- Paper-specific scope condition.
- Paper-stated limitation.
- Transferability to SSA tax earmarking.
- Verification flag.

10.2 Core evidence matrix

The core matrix shows that direction varies by fiscal stage and that most positive findings concern authorization or gross receipts rather than net SSA DRM.

Source	Focal claim	Evidence type	Setting	Direction and outcome	Paper-specific scope condition
Buchanan (1963)	Earmarking permits item-by-	Formal public-choice model	Abstract; US illustrations	Conditional for allocation and total	Median voter, fixed tax price, negligible

Source	Focal claim	Evidence type	Setting	Direction and outcome	Paper-specific scope condition
	item choice; general-fund effects depend on budget ratios and elasticities			outlay; DRM not measured	decision cost, binding execution
Lee & Wagner (~1991)	Quasi-pricing can improve choice, but discriminatory politics can turn earmarking into rent extraction	Normative comparative-institutional argument	Advanced/US	Mixed or indeterminate	Complementarity, uniform payer–benefit relation, constitutional non-discrimination
Teja (1988)	Earmarking can address preference, bureaucracy, commitment, and trust problems	Formal/normative analysis plus descriptive case	Other LMIC; Colombia	Conditional; compliance untested	Participation, review, benefit link, limited scale, joint finance
McCleary (1991)	Efficiency and automaticity are usually incompatible; local strong earmarks perform better	Operational review plus analytical decomposition	Mixed; four SSA road cases	Mostly negative nationally; positive local valorization	Payer–beneficiary overlap, private-good traits, fiscal commitment
Bös (2000)	Asymmetric information reduces optimal earmarking; political earmarking can raise cost	Formal principal–agent model	Abstract; advanced motivation	Negative for efficiency under modeled politics	Tax-before-spend timing, separate taxpayer/spender, political weights
Brett & Keen (2000)	Credible earmarking protects revenue from successor expropriation and supports a higher tax	Formal signalling model	Abstract	Positive for tax level; conditional for welfare	Credible commitment; political uncertainty; earmarking cost below expropriation risk
Marsiliani & Renström (2000)	Binding earmarking partly mitigates time inconsistency but cannot restore commitment optimum	Formal dynamic model	Abstract	Positive but partial	Constitutionally binding rule; inequality and tax-base structure
Flores-Macías (2018)	Exclusive-use wording raises stated tax support	Randomized survey experiment	Other LMIC; Mexico	Positive for stated support	Hypothetical public-safety proposal; strongest among low-trust respondents

Source	Focal claim	Evidence type	Setting	Direction and outcome	Paper-specific scope condition
Prichard (2009; 2015)	Ghanaian earmarks helped secure contentious tax measures	Process tracing; 58 interviews and documents	SSA; Ghana	Positive for passage; mixed for additionality and governance	Salience, organized bargaining, oversight, pre-emptive adoption
Cashin et al. (2017)	Health-earmark outcomes depend on design and PFM; additionality is uncertain	Operational review and six-country case comparison	Mixed; Ghana and South Africa in SSA	Positive for some adoption/programs; mixed fiscal results	Health–finance alignment, PFM, absorptive capacity, flexible linkage
WHO (2016)	Tobacco earmarks can be enacted and administered through several fund models	Selected descriptive case series	Mixed; Botswana in SSA	Positive for political/administrative feasibility; outcomes not established	Enabling regulation, transfer mechanism, governance, generally small scale
Kaiser et al. (2016)	Philippine earmarking formed part of a broader reform coalition	Process tracing	Other LMIC; Philippines	Positive for reform coalition; mixed execution	Political leadership, annual budgeting, targeting, implementation capacity
Kimwell et al. (2022)	Philippine appropriations rose, but counterpart substitution and implementation weaknesses persisted	Accounting review, ARMAX/ARDL, interviews	Other LMIC; Philippines	Mixed	Soft attribution, short time series, procurement and local capacity
Gwilliam & Shalizi (1999)	Second-generation road funds can improve maintenance management in an intermediate governance range	Operational policy review	SSA-heavy mixed sample	Conditional positive	User-charge relation, user board, audit, additional revenue, macrofiscal control
Bird & Jun (2005)	Benefit link and marginal bite determine economic substance	Taxonomy, evidence review, accounting case	Advanced/other; Korea	Mixed	Marginal source observable; strong link; descriptive case only
Allen & Radev (2006)	EBF benefits depend on governance and consolidation; poorly managed funds undermine PFM	IMF operational and institutional review	Mixed, including SSA examples	Conditional; system risks usually negative under weak PFM	Consolidation, common controls, audit, integrated collection, limited proliferation
Stivender et al.	Education earmarking may	Staggered difference-in-differences	Advanced; US states	Weakly positive and specification-sensitive	Voluntary purchase, few legal changes,

Source	Focal claim	Evidence type	Setting	Direction and outcome	Paper-specific scope condition
(2016)	increase lottery sales				beliefs unobserved
Jones (2013)	Education lotteries raise non-tax revenue without clear education-spending additionality	Difference-in-differences/accounting decomposition	Advanced; US states	Positive revenue; zero education spending; negative private giving	Lottery setting, substitution across budgets and donations
Blanchet et al. (2012)	Ghana NHIS membership is associated with higher formal care use	Propensity-score matching observational study	SSA; Ghana	Positive for utilization	Selection into insurance; financing rule not identified
Moore, Prichard & Fjeldstad (2018)	Selective earmarking may form part of a low-trust African fiscal contract	Comparative descriptive synthesis	SSA	Conditional	Salience, transparency, monitoring, political bargaining
Wilkinson (1994)	Strong/narrow earmarks are more transparent; weak earmarks can mislead	Normative policy analysis	Advanced	Conditional	Strength and breadth of legal link; procyclical risk
Ahmad & Brosio (2008)	Earmarked transfers can weaken local effort and accountability	Comparative institutional analysis	Other LMIC; Latin America	Negative for transfer-dependent local effort	Intergovernmental dependence; not national tax hypothecation
Smart & Bird (2009)	Grant earmarks can target need, harden budgets, or alter accountability, with fungibility varying by capacity	Conceptual review with documentary evidence	Advanced plus Colombia	Mixed	Recipient own-source capacity, information asymmetry, equalization, salience
Luttmer & Singhal (2014)	Reciprocity is one tax-morale mechanism, but revenue-use messages usually do not move compliance	Structured evidence review	Mostly advanced; some LMIC	Mechanism plausible; interventions mostly zero	Belief strength, salience, outcome margin, enforcement
Flores-Macías & Sánchez-Talanquer	Perceived national state failure is associated with lower willingness to pay	Cross-sectional interval regression	Other LMIC; Mexico	Negative association	Public-safety setting; no earmark; no causal identification

10.3 Adjacent, methodological, or excluded entries

Adjacent and methodological sources require separate rows so that coding, grant, tax-morale, and general DRM findings are not read as direct evidence on earmarked taxes.

Source	Use in this review	Evidence type	Setting	Direction or role	Limitation
Wagner (1991)	Consent-of-the-taxed and constitutional principle/practice gap	Normative argument	Advanced/US	Conditional legitimacy	Consent not directly observable; tobacco-industry funding provenance
Fonseca Corona (2024)	Documents methodological concentration and single-mechanism gap	Scoping review	Mixed	Gap identification	No pooled effects; no earmarking study
Das et al. (2026)	Coding architecture and adjacent implementation-friction mechanism	LLM narrative database; proxy-SVAR/LP-IV	Mixed; nine SSA countries	Methods-positive; substantive analogy only	No earmarking; policy stages mixed; political state not randomized
Chelsky (2023)	Demonstrates tax-policy reversal and election timing outside earmarking	Evaluation synthesis/blog-style discussion	Other LMIC	Negative for durability of some reforms	No earmarking comparison; illustrative evidence
OECD/AUC/ATAF (2025)	Shows standard revenue classification omits destination	Statistical manual/report	Africa	Measurement role	No causal evidence; contextual statistics excluded; no IMF-equivalent classification statement in corpus
Mankiw, Weinzierl & Yagan (2009)	General-fund optimal-tax benchmark	Normative/theoretical review	Advanced/abstract	No earmarking result	Benefit principle largely outside its scope
Akitoby et al. (2019)	Excluded to the introduction	Comparative DRM case studies	LMIC	General DRM context	Does not establish an earmarking effect
Chelsky (2023), budget support	Excluded except for general attribution caution	Evaluation discussion	LMIC/fragile states	Fungibility/traceability analogy	Budget support, not earmarked taxation
Smart & Bird (2009), CHST item	Illustrates statutory form versus fiscal behavior in a grant	Documentary comparison	Advanced/Canada	Mixed	[VERIFY: CHST expansion and the 1995 conversion/creation date against the original paper.]

11. Author's synthesis and bridge to the conceptual framework

Author's conceptual finding: Earmarking can support DRM only if a gain at authorization or compliance survives every downstream gate—net collection, executed transfer, marginal expenditure, expenditure quality, and budget-wide control.

11.1 Derived channels

The transmission chain yields eight distinct channels whose signs and evidentiary status must be evaluated separately.

- **Authorization channel:** A visible expenditure link can make a tax proposal easier to enact by creating a coalition, lowering opposition, or allowing organized taxpayers to exchange acceptance for a specified benefit; Prichard, Flores-Macías, Cashin et al., Kaiser et al., and Allen and Radev most directly motivate this channel.
- **Compliance channel:** A credible benefit or reciprocity signal may change voluntary payment only where taxpayers are aware of the link and possess a meaningful evasion, reporting, payment, or political-refusal margin; Teja and Luttmer and Singhal motivate the mechanism, but the corpus does not identify it for an SSA earmark.
- **Benefit-link channel:** Payer–beneficiary overlap can make an earmarked tax approximate a price and improve demand signalling; Lee and Wagner, McCleary, Teja, Bird and Jun, and the road-fund literature motivate this channel.
- **Commitment channel:** A legally durable earmark can constrain future diversion, successor expropriation, or time-inconsistent taxation; Brett and Keen, Marsiliani and Renström, Teja, and Bird and Jun motivate this channel.
- **Budget-additionality channel:** Statutory revenue changes target spending only if the transfer is executed and remains the marginal source after general-fund offsets, other transfers, and private contributions are considered; Bird and Jun, Cashin et al., Prichard, Kimwell et al., Jones, and McCleary motivate this channel.
- **Expenditure-quality channel:** Additional resources improve outcomes only if procurement, staffing, purchasing, targeting, governance, audit, and absorptive capacity convert them into useful services; Allen and Radev, Gwilliam and Shalizi, Cashin et al., Kaiser et al., Kimwell et al., and Blanchet et al. motivate this channel.
- **System-cost channel:** Multiple earmarks can fragment allocation, cash management, tax administration, reporting, borrowing control, and political accountability even where an individual fund appears successful; Allen and Radev and McCleary most directly motivate this channel.
- **Feedback channel:** Executed and visible service improvements may strengthen future trust and willingness to pay, while broken promises, diversion, poor quality, or repeal may weaken the credibility of subsequent earmarks; Luttmer and Singhal's persistence mechanism and the de jure/de facto cases motivate this dynamic extension.

11.2 Conditions under which earmarking could support DRM

Earmarking supports DRM only when the political or compliance gain is incremental, fiscally net-positive, executed, additional, usable, governable, and compatible with comprehensive budgeting.

- **Incremental political feasibility:** The associated tax must be less likely to pass, survive, or maintain its rate without the earmark; simple co-occurrence with a popular expenditure program is insufficient.
- **Relevant taxed population:** Political approval should come from or credibly include those bearing the tax rather than principally from a majority taxing a politically weak minority.
- **Meaningful compliance margin:** Taxpayers must be able to alter reporting, payment, participation, or political resistance; taxes fully withheld or invisibly embedded may have a smaller individual reciprocity margin.
- **Credible dedication:** The legal rule, implementing regulation, transfer mechanism, appropriation process, and amendment regime must make the promised destination believable.
- **Net revenue gain:** Additional gross receipts must exceed base erosion, administrative cost, displacement of other taxes, reduced profits or consumption, exemptions, nonpayment by public entities, and future rate reversal.
- **Marginal budget bite:** Executed earmarked revenue must increase target spending or improve its composition rather than merely replace general-fund, donor, subnational, or private financing.
- **Expenditure usability:** The target sector must possess the capacity to procure, spend, monitor, and evaluate the additional resources without persistent balances, low execution, cost inflation, or program proliferation.
- **Governance quality:** The fund or account must have clear responsibility, purchaser discipline where relevant, complete accounts, legislative scrutiny, independent audit, transparent reporting, and protection from producer capture.
- **Budget compatibility:** The earmark must remain inside consolidated fiscal reporting, the treasury and cash-management framework, commitment controls, debt analysis, and medium-term prioritization.
- **Limited rigidity:** Review clauses, sunset provisions, release valves, and explicit general-fund supplementation should permit adjustment when the revenue base, program need, or relative policy priority changes.
- **Limited proliferation:** The gain from one dedicated arrangement must be assessed against the precedent it creates for other sectors and the cumulative loss of comprehensive budgeting.

- **Distributional legitimacy:** The tax base, financed benefit, and incidence should not rely principally on discriminatory extraction or hostility toward a narrowly defined group.
- **Resource-fund separation:** Saving and stabilization of non-tax resource revenue should not be counted as evidence that an earmarked tax mobilizes resources for current expenditure.
- **Durability evidence:** Claims that earmarking protects a tax against reversal require direct comparison with otherwise similar non-earmarked taxes; the current corpus does not provide it.

11.3 SSA institutional prior

The author's SSA prior is strongest for a short-run authorization effect, weaker for a compliance effect, conditional for budget additionality, and skeptical about expenditure-quality and system-wide gains without evidence on governance and execution.

- The corpus supports a "demand–enforceability paradox" as a prior, not as an established empirical regularity.
- Teja, Brett and Keen, Marsiliani and Renström, Flores-Macías, Moore et al., and Prichard place the demand for visible dedication in settings characterized by mistrust, political weakness, inequality, resistance, or poor ordinary budgeting.
- Lee and Wagner, McCleary, Gwilliam and Shalizi, Cashin et al., and Allen and Radev place successful operation under conditions of credible non-discrimination, enforceable commitment, close payer–beneficiary linkage, capable PFM, audit, tax administration, and expenditure capacity.
- Several required conditions may therefore be least secure precisely where political demand for earmarking is strongest.
- The Ghana cases show that the paradox is not deterministic: earmarking can facilitate passage and produce early expenditure additionality even where longer-run substitution, bailout, corruption, or system fragmentation remain concerns.
- The African road-fund evidence similarly suggests an intermediate institutional range rather than a monotonic governance result.
- The prior for SSA should therefore be strongest for a short-run authorization effect, weaker for a compliance effect, conditional for budget additionality, and explicitly skeptical about expenditure quality and system-wide gains without evidence on governance and execution.

11.4 Transition to the conceptual framework

The conceptual framework should treat de jure design as the first node in a sequence of separately measurable gaps rather than as a proxy for behavior.

- **Framework chain:** De jure design → awareness and perceived credibility → political authorization → collection and tax-base response → transfer and release → marginal sectoral expenditure and composition → expenditure quality and service delivery → trust, compliance, and policy durability.
- **Design moderators:** Tax visibility; who remits; base elasticity; payer–beneficiary overlap; discriminatory incidence; legal tier; implementing regulation; appropriation discretion; on/off-budget location; transfer automaticity; amendment and sunset rules.
- **Institutional moderators:** Legislative cohesion; electoral timing; political competition; civil-society monitoring; tax-administration capacity; PFM credibility; treasury integration; external audit; fund-board structure; fiscal stress; donor dependence; subnational own-source capacity.
- **Sector moderators:** Divisibility; returns to scale; externalities; expenditure lumpiness; maintenance intensity; absorptive capacity; procurement complexity; geographic targeting; ability to identify beneficiaries.
- **Distinct estimands:** Probability of enactment; statutory rate; effective rate; gross receipts; net general-government revenue; executed transfer; marginal target spending; recurrent/capital and wage/non-wage composition; balances and borrowing; output cost; service use; distribution; repeal or suspension hazard.
- **Measurement sequence:** A de jure inventory identifies the institutional treatment and risk set; execution and accounts establish whether the rule operated; marginal-spending analysis establishes additionality; output and outcome analysis establishes expenditure performance.
- **Principal empirical proposition:** Earmarking supports DRM only when its incremental passage or compliance gain exceeds tax distortion and administrative cost, survives the budget process, produces net rather than displaced revenue, and finances expenditure that can be used effectively within a comprehensive fiscal framework.
- **Null proposition:** An earmark can increase political support while leaving compliance, net revenue, target spending, or service quality unchanged.
- **Adverse proposition:** An earmark can make a tax politically feasible while reducing aggregate revenue, institutionalizing discrimination, weakening budget comprehensiveness, or directing resources into a poorly governed fund.
- **Contribution of de jure coding:** Statutory form is not the behavioral outcome; it is the necessary first node that permits the subsequent gaps—announcement versus law, law versus regulation, regulation versus transfer, transfer versus spending, and spending versus quality—to be measured rather than assumed.